

Karnataka District Development Index

A Data-Driven Analysis of 31 Districts Using Machine Learning and Socioeconomic Indicators

Author: Rohan Shetty | Qualification: MSc Business Analytics

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Tools Used: Python | SQL | Power BI | Machine Learning

Data Source: Karnataka Economic Survey 2025-26 | data.opencity.in

This study presents a comprehensive data-driven analysis of socioeconomic development across 31 districts of Karnataka using seven government datasets from 2024-26. A composite District Development Index (DDI) was constructed using weighted indicators spanning income, employment, education, housing, and informal economy.

Machine learning models including K-Means Clustering, Random Forest Regression, Random Forest Classification, and OLS Regression were applied to predict and classify district development levels. Results reveal extreme regional inequality — Bengaluru Urban scores 79.79 while Kalaburagi scores only 9.82 on the DDI. Regional analysis confirms that Coastal Karnataka consistently outperforms North Karnataka across all indicators. Time series analysis reveals a statewide decline in education performance between 2024 and 2025. Gini coefficient analysis shows that while educational access is relatively equal (Gini 0.04), employment through MSMEs is highly concentrated (Gini 0.40). Policy interventions targeting education and income in North Karnataka are strongly recommended.

Keywords: District Development Index, Karnataka, Machine Learning, K-Means Clustering, Random Forest, Socioeconomic Analysis, Regional Inequality

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1. Introduction

Karnataka is the third largest economy in India with a Gross State Domestic Product (GSDP) of approximately ₹33 trillion as of 2026. The state is home to Bengaluru Urban — India's technology capital — which alone contributes nearly 40% of Karnataka's total economic output. However, this remarkable economic growth has not been uniformly distributed across the state's 31 districts.

A significant development divide exists between the prosperous southern and coastal districts and the historically underdeveloped northern districts. Districts such as Kalaburagi, Yadagiri, and Raichur in North Karnataka continue to lag behind in income, education, employment, and basic infrastructure — despite decades of government intervention and policy attention.

Traditional approaches to measuring district development often rely on single indicators such as per capita income or literacy rate. These single-indicator approaches fail to capture the multidimensional nature of development. A composite index that combines multiple socioeconomic indicators provides a more accurate and holistic picture of district-level development.

This study addresses this gap by constructing a District Development Index (DDI) for all 31 Karnataka districts using seven government datasets from 2024-26. Advanced machine learning models are applied to predict and classify development levels, and regional analysis is conducted to identify systematic patterns of

inequality across Karnataka's five geographic regions.

1.1 Research Objectives

This study has three primary objectives:

1. To analyse socioeconomic indicators across all 31 Karnataka districts and identify patterns of development and underdevelopment.
2. To construct a composite District Development Index using weighted socioeconomic indicators and rank all 31 districts.
3. To apply Machine Learning models to predict district development levels and identify the most significant drivers of district development in Karnataka.

1.2 Research Questions

This study seeks to answer three research questions:

1. Which districts of Karnataka are most and least developed, and what socioeconomic patterns characterise each group?
2. Which indicators have the strongest influence on district development in Karnataka?
3. Can machine learning models accurately predict and classify district development levels based on socioeconomic indicators?

1.3 Significance of the Study

This study makes three important contributions:

First, it provides the first comprehensive machine learning based district development analysis for Karnataka using the most recent 2024-26 government data.

Second, the District Development Index developed in this study can serve as an annual monitoring tool for

policymakers to track district-level progress over time.

Third, the predictive models developed allow simulation of policy scenarios — enabling policymakers to estimate the impact of specific interventions before implementing them.

2. Data and Methodology

2.1 Data Sources

This study uses seven district-level datasets published by the Government of Karnataka through the Open City Data Portal (data.opencity.in). All datasets are from the Karnataka Economic Survey 2025-26 and related government portals.

Dataset 1: District-wise GDDP and Per Capita Income

Source: Karnataka Economic Survey 2025-26

Coverage: 31 districts

Key variables: Per Capita Income (Rs.), GDDP at Current Prices (Cr.)

Dataset 2: District-wise MSME Industries and Employment Generated

Source: Karnataka Economic Survey 2025-26

Coverage: 31 districts

Key variables: Total Enterprises, Total Employed

Dataset 3: District-wise Housing Completed

Source: Karnataka Economic Survey 2025-26

Coverage: 31 districts

Key variables: Houses completed under various government schemes

Dataset 4: District-wise Registered in

Yuva Nidhi Scheme

Source: Karnataka Economic Survey 2025-26

Coverage: 31 districts

Key variables: Total unemployed youth registered

Dataset 5: District-wise Number Registered
under eShram Portal

Source: Karnataka Economic Survey 2025-26

Coverage: 31 districts

Key variables: Registration achievement,

Coverage percentage

Dataset 6: District-wise Number Passed SSLC
2024 and 2025

Source: Karnataka Economic Survey 2025-26

Coverage: 31 districts

Key variables: Pass percentage 2024, Pass
percentage 2025

Dataset 7: District-wise PUC Pass Percentage
2024 and 2025

Source: Karnataka Economic Survey 2025-26

Coverage: 31 districts

Key variables: Pass percentage 2024, Pass
percentage 2025

2.2 Data Preprocessing

The raw datasets presented significant preprocessing challenges. The same district appeared with different spellings across datasets — for example Bangalore Urban, Bengaluru Urban, and Bengaluru (Urban) all referring to the same district. A comprehensive name standardization dictionary with 35 mapping entries was developed to resolve these inconsistencies.

Key preprocessing steps:

- Standardized 50+ district name variations across all 7 datasets
- Removed duplicate rows created by sub-district splits using groupby aggregation

- Imputed missing districts using column mean values
- Converted string columns to numeric format
- Removed aggregate rows (Total, State level rows)

Final master dataset: 31 districts × 22 indicators

Missing values after preprocessing: 0

2.3 District Development Index Construction

The District Development Index was constructed using a three-step process:

Step 1 — Normalization:

All indicators were normalized to a 0-1 scale using Min-Max normalization to remove the effect of different units and scales.

Step 2 — Direction Adjustment:

For negative indicators where higher values indicate lower development, the normalized score was reversed:

Reversed Score = 1 - Normalized Score

Negative indicators: eShram Coverage % and Unemployed Youth

Step 3 — Weighted Scoring: A weighted sum was calculated using the following weights based on theoretical importance:

Per Capita Income = 25%

Total Employed = 20%

SSLC Pass Average = 15%

PUC Pass Average = 15%

eShram Coverage % = 10%

Unemployed Youth = 10%

Total Houses = 5%

Total = 100%

Final DDI Score = Weighted Sum × 100

Range: 0 to 100

2.4 Machine Learning Models

Four machine learning models were developed:

Model 1 — K-Means Clustering (Unsupervised):

Purpose: Group districts into development tiers

Input: 7 normalized socioeconomic indicators

Optimal K: 3 (selected using Elbow method)

Output: Three tiers — Highly Developed, Developing, Underdeveloped

Model 2 — Random Forest Regression (Supervised):

Purpose: Predict DDI Score

Input: 7 socioeconomic indicators

Output: Continuous DDI Score (0-100)

Evaluation: R^2 Score, MAE, 5-fold Cross Validation

Model 3 — Random Forest Classification (Supervised):

Purpose: Predict Development Tier

Input: 7 socioeconomic indicators

Output: Development Tier label

Evaluation: Precision, Recall, F1-Score

Model 4 — OLS Regression (Statistical):

Purpose: Identify statistically significant drivers of development

Input: Standardized socioeconomic indicators

Output: Regression coefficients and p-values

Evaluation: R^2 , Adjusted R^2 , F-statistic

2.5 Advanced Analysis Methods

Time Series Analysis:

Year-on-year change in SSLC and PUC pass rates between 2024 and 2025 was calculated to identify districts with improving or declining education performance.

Regional Analysis:

31 districts were grouped into 5 geographic regions:

- Coastal Karnataka (3 districts)
- Malnad Karnataka (2 districts)
- South Karnataka (11 districts)
- Central Karnataka (2 districts)
- North Karnataka (13 districts)

Gini Coefficient Analysis:

Gini coefficients were calculated for six key indicators to measure inequality of distribution across districts.

Sensitivity Analysis:

DDI scores were recalculated under three different weight scenarios to test the robustness of rankings.

Future Prediction Scenarios:

The Random Forest model was used to predict DDI scores for bottom 5 districts under three improvement scenarios — 10% income increase, 10% education improvement, and combined improvement.

2.6 Tools and Technologies

Programming Language = Python 3.12

Data Manipulation = Pandas, NumPy

Machine Learning = Scikit-learn

Statistical Analysis = Statsmodels

Visualization = Matplotlib, Seaborn

Database = SQLite3

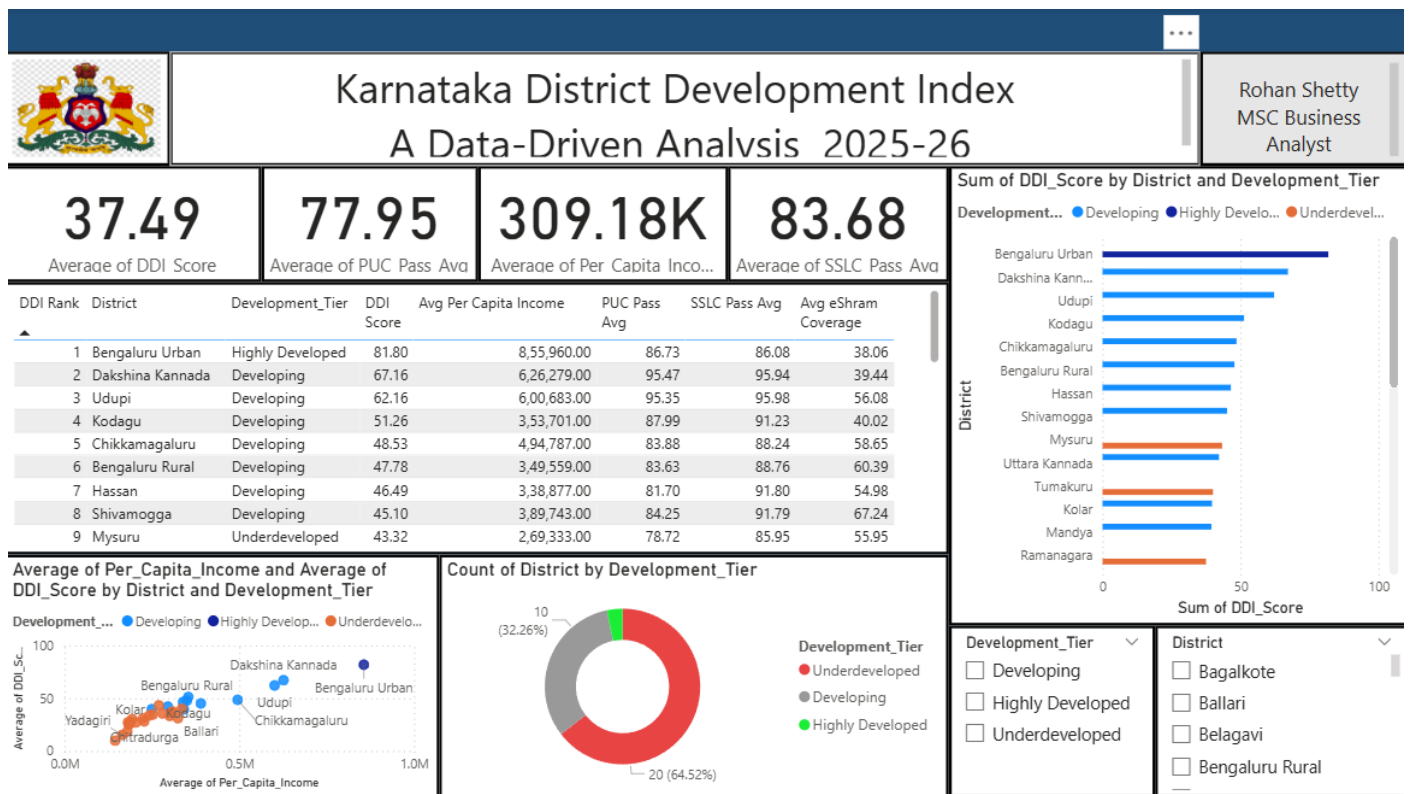
Dashboard = Microsoft Power BI

Environment = Jupyter Notebook(Anaconda)

Karnataka District Development

Index — Power BI Interactive Dashboard

The interactive dashboard was developed in Microsoft Power BI Desktop to visualize the District Development Index findings. The dashboard includes KPI cards showing state averages, a ranked bar chart of all 31 districts by DDI Score, a donut chart showing development tier distribution, a scatter plot of income vs DDI Score, and an interactive data table. Slicers enable dynamic filtering by Development Tier and District.



3. Results and Findings

3.1 Descriptive Statistics

Analysis of the master dataset reveals significant variation across Karnataka's 31 districts:

Per Capita Income:

- State average: ₹3,09,185
- Highest: Bengaluru Urban ₹8,55,960
- Lowest: Kalaburagi ₹1,44,449
- Bengaluru Urban is 6 times wealthier than Kalaburagi — revealing extreme income inequality

SSLC Pass Rate:

- State average: 80.99% (2025)
- Highest: Udupi 93.94%
- Lowest: Kalaburagi 56.89%
- Gap of 37 percentage points between best and worst performing district

PUC Pass Rate:

- State average: 72.15% (2025)
- Highest: Udupi 93.90%
- Lowest: Kalaburagi 55.70%
- Nearly half of students in bottom districts are failing PUC examinations

eShram Coverage:

- State average: 64.57%
- Highest: Vijayapura 88.47%
- Lowest: Bengaluru Urban 38.06%
- Higher coverage indicates more informal workers and lower development

3.2 Correlation Analysis

Correlation analysis reveals four key relationships:

Finding 1 — Economy and Employment are inseparable:

GDDP and Total MSME Units show a correlation of 0.97

GDDP and Total Employed show a correlation of 0.94

Interpretation: Economic growth and job creation are almost perfectly linked in Karnataka districts.

Finding 2 — Income drives education:

Per Capita Income and PUC Pass Rate show a correlation of 0.74

Per Capita Income and SSLC Pass Rate show a correlation of 0.60

Interpretation: Wealthier districts consistently achieve better educational outcomes.

Finding 3 — Informal economy signals underdevelopment:

eShram Coverage and Per Capita Income show a correlation of -0.69

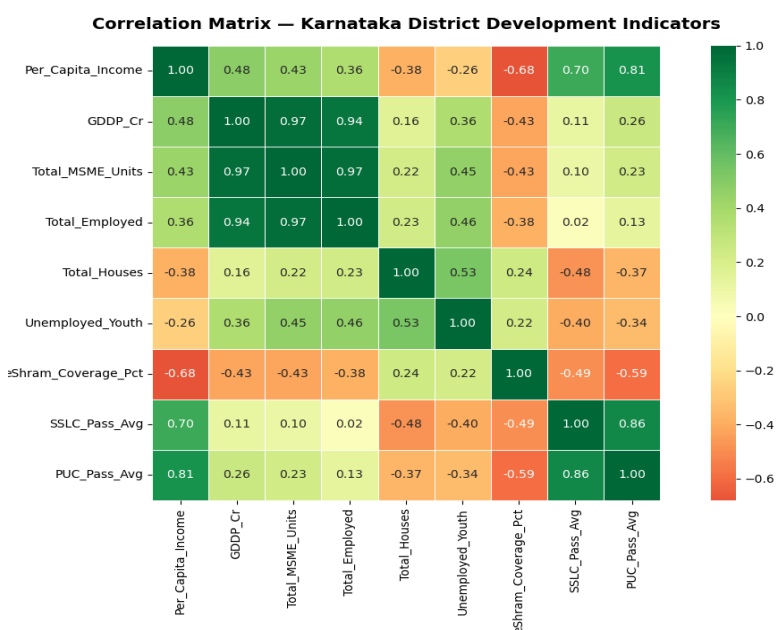
eShram Coverage and PUC Pass Rate show a correlation of -0.59

Interpretation: Districts with more informal workers have lower income and worse education.

Finding 4 — Education is internally consistent:

SSLC Pass Rate and PUC Pass Rate show a correlation of 0.86

Interpretation: Districts that perform well at Class 10 level also perform well at Class 12 suggesting systemic education quality differences across districts.



Key Fact: GDP and MSME Employment shows correlation of 0.97 — The strongest relationship in the entire dataset

3.3 District Development Index Results

The District Development Index scores reveal a clear hierarchy of development:

Top 5 Most Developed Districts:

Rank 1: Bengaluru Urban = DDI Score: 79.79

Rank 2: Dakshina Kannada = DDI Score: 75.23

Rank 3: Udupi = DDI Score: 69.80

Rank 4: Kodagu = DDI Score: 54.77

Rank 5: Chikkamagaluru = DDI Score: 54.40

Bottom 5 Least Developed Districts:

Rank 27: Bidar = DDI Score: 25.92

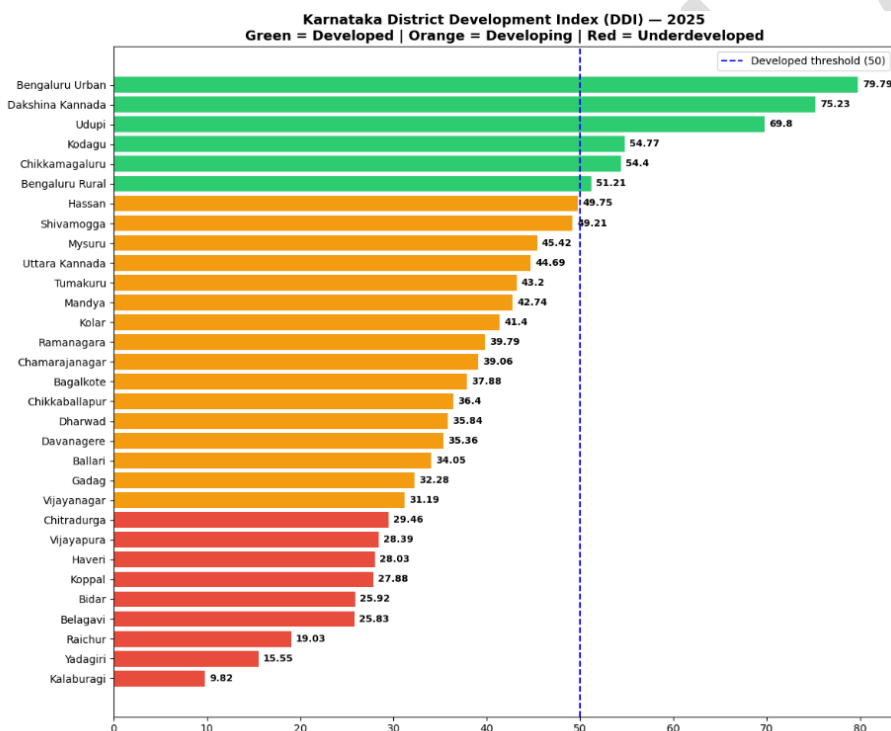
Rank 28: Belagavi = DDI Score: 25.83

Rank 29: Raichur = DDI Score: 19.03

Rank 30: Yadagiri = DDI Score: 15.55

Rank 31: Kalaburagi = DDI Score: 9.82

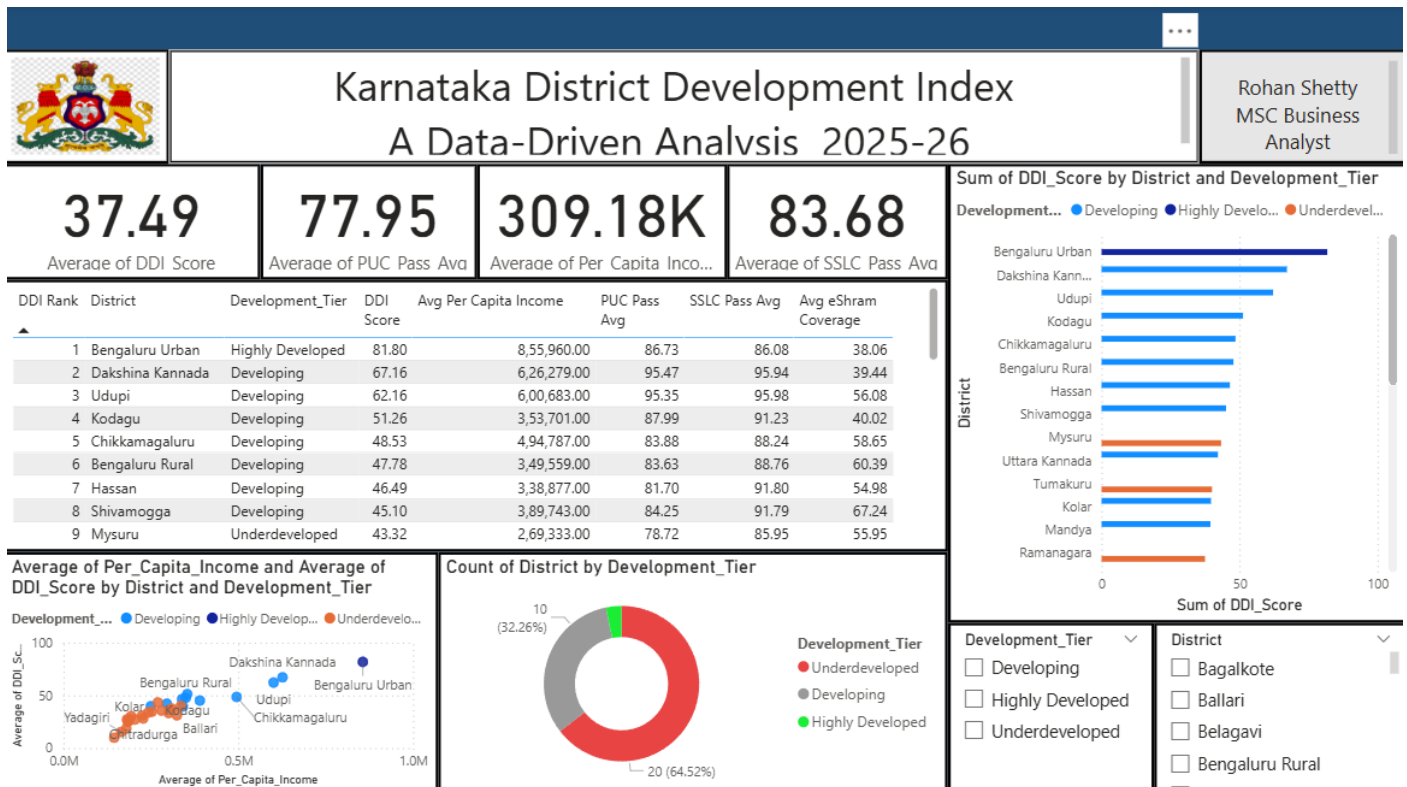
The gap between the highest scoring district (Bengaluru Urban 79.79) and the lowest scoring district (Kalaburagi 9.82) is 69.97 points representing an extreme development divide within a single state.



Key Fact: The gap between Bengaluru Urban (79.79) and Kalaburagi (9.82) is 69.97 points — nearly 7 times difference in development score

Power BI Dashboard — DDI Score Rankings and Development Tier Distribution

The Power BI dashboard provides an interactive view of the District Development Index results. The dashboard confirms that Bengaluru Urban leads with a DDI Score of 79.79 while Kalaburagi scores only 9.82. The donut chart shows that 64.52% of Karnataka districts fall in the Underdeveloped tier.



3.4 K-Means Clustering Results

K-Means clustering with K=3 identified three distinct development tiers:

Tier 1 — Highly Developed (1 district, 3%):Bengaluru Urban

Average DDI Score: 79.79

Tier 2 — Developing (10 districts, 32%):Dakshina Kannada, Udupi, Kodagu, Chikkamagaluru, Bengaluru Rural, Hassan, Shivamogga, Uttara Kannada,Kolar, Mandya

Average DDI Score: 53.32

Tier 3 — Underdeveloped (20 districts, 65%):All remaining 20 districts

Average DDI Score: 31.02

Critical finding: 65% of Karnataka districts are classified as Underdeveloped. Only 1 district achieves Highly Developed status. This confirms that Karnataka's development is heavily concentrated in a single urban center.

MODEL 3 – RANDOM FOREST CLASSIFICATION

Classification Report:

	precision	recall	f1-score	support
Developing	1.00	1.00	1.00	10
Highly Developed	1.00	1.00	1.00	1
Underdeveloped	1.00	1.00	1.00	20
accuracy			1.00	31
macro avg	1.00	1.00	1.00	31
weighted avg	1.00	1.00	1.00	31

Model 4 — OLS Regression:

$$R^2 = 1.000$$

$$\text{Adjusted } R^2 = 1.000$$

$$\text{F-statistic} = 1.306 \times 10^8$$

$$\text{All p-values} < 0.001$$

Regression Coefficients (standardized):

Per Capita Income = +5.325 (strongest positive)

PUC Pass Average = +3.600

SSLC Pass Average = +3.445

Total Employed = +3.425

Total Houses = +1.268

Unemployed Youth = -2.152

eShram Coverage % = -2.468 (strongest negative)

Interpretation: Per Capita Income has the strongest positive effect on development. eShram Coverage has the strongest negative effect. All coefficients are statistically significant at $p < 0.001$.

```

MODEL 4 - OLS REGRESSION
=====
                        OLS Regression Results
=====
Dep. Variable:          DDI_Score      R-squared:                1.000
Model:                  OLS            Adj. R-squared:           1.000
Method:                 Least Squares   F-statistic:              9.625e+07
Date:                   Thu, 01 Oct 2026 Prob (F-statistic):       2.60e-84
Time:                   10:44:33        Log-Likelihood:           137.17
No. Observations:       31             AIC:                     -258.3
Df Residuals:           23             BIC:                     -246.9
Df Model:                7
Covariance Type:        nonrobust
=====
                        coef      std err      t      P>|t|      [0.025      0.975]
-----
const                39.7861      0.001    6.58e+04    0.000      39.785      39.787
Per_Capita_Income     6.5109      0.001    5103.734    0.000      6.508      6.514
Total_Employed         3.4244      0.001    3797.048    0.000      3.423      3.426
Total_Houses           1.2685      0.001    1614.054    0.000      1.267      1.270
Unemployed_Youth       -2.1525      0.001   -2504.108    0.000      -2.154      -2.151
eShram_Coverage_Pct    -2.4682      0.001   -2809.222    0.000      -2.470      -2.466
SSLC_Pass_Avg          3.4453      0.001    2696.230    0.000      3.443      3.448
PUC_Pass_Avg           3.5983      0.002    2369.340    0.000      3.595      3.601
=====
Omnibus:                6.565    Durbin-Watson:           2.418
Prob(Omnibus):          0.038    Jarque-Bera (JB):        2.132
Skew:                   -0.185    Prob(JB):                0.344
Kurtosis:                1.769    Cond. No.                6.08
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

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3.6 Feature Importance

Random Forest feature importance analysis reveals:

Per Capita Income = 60.6%

PUC Pass Average = 17.3%

SSLC Pass Average = 14.6%

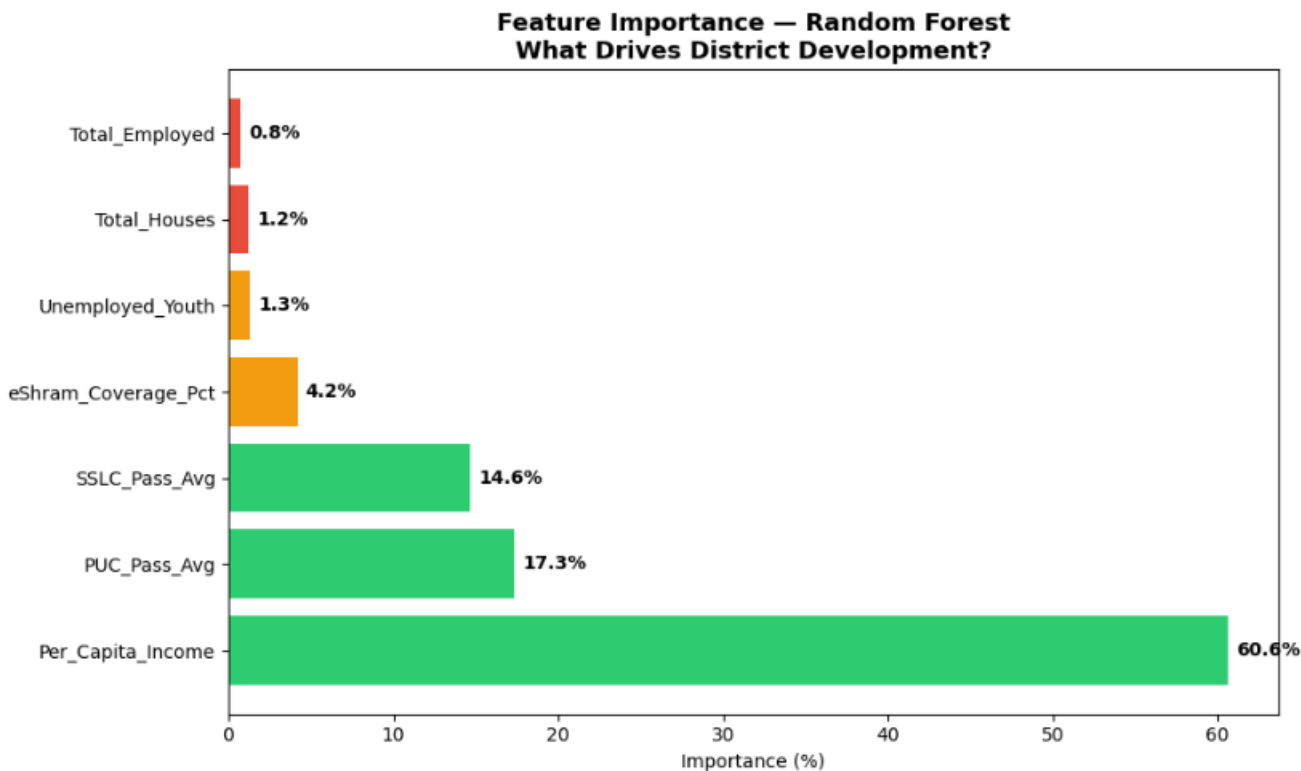
eShram Coverage % = 4.2%

Unemployed Youth = 1.3%

Total Houses = 1.2%

Total Employed = 0.8%

Income alone drives 60.6% of development prediction. Education drives another 31.9%. Together they explain 92.5% of district development variation.



3.7 Time Series Analysis — Education Trend

Comparison of SSLC and PUC pass rates between 2024 and 2025 reveals an alarming trend:

SSLC Performance:

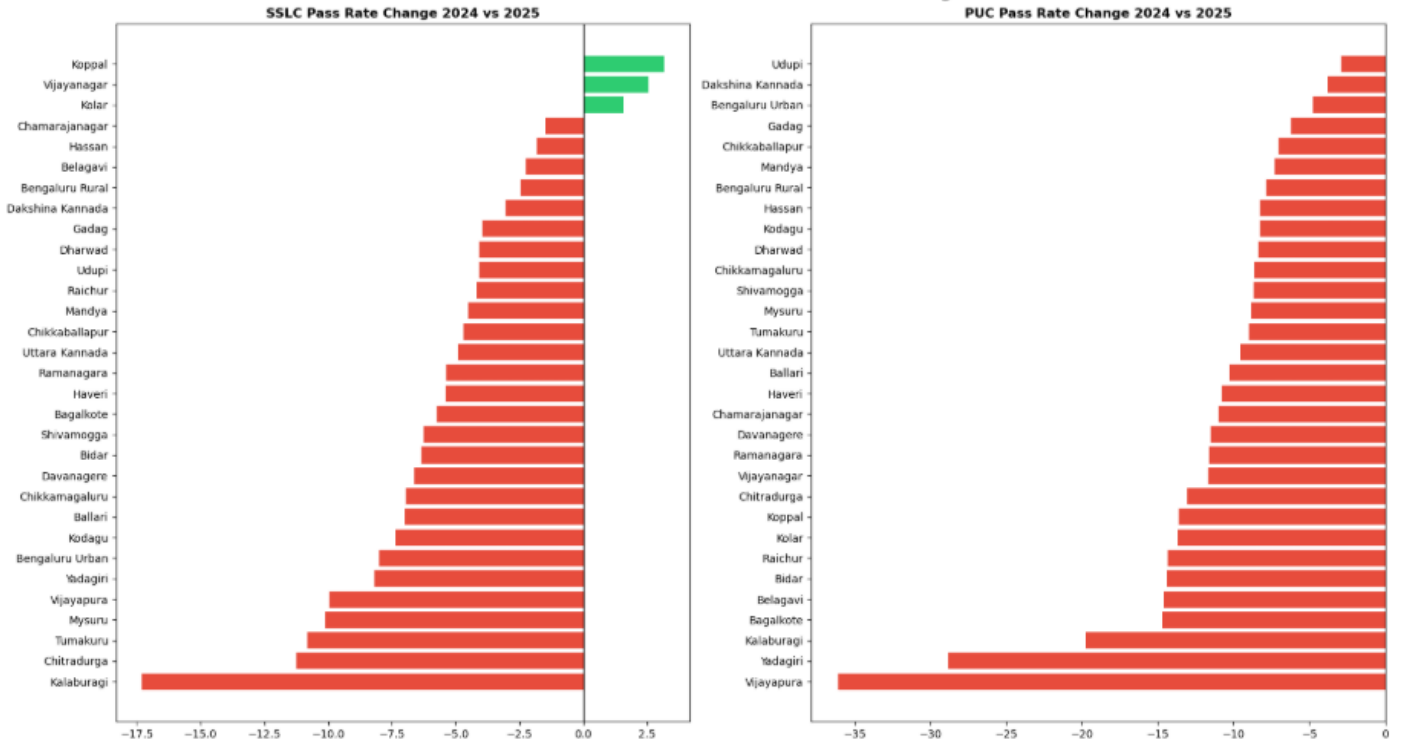
- Only 3 of 31 districts improved in 2025
- Koppal: +3.16%, Vijayanagar: +2.55%, Kolar: +1.57%
- 28 districts declined in SSLC pass rate
- Worst decline: Kalaburagi -17.29%

PUC Performance:

- All 31 districts declined in PUC pass rate
- Worst decline: Vijayapura -36.08%
- Second worst: Yadagiri -28.84%

This universal decline suggests either a genuine deterioration in education quality or a change in examination standards between 2024 and 2025. Further investigation is recommended.

Karnataka Education Performance — Year on Year Change



Key Fact: All 31 districts declined in PUC pass rate in 2025. Vijayapura recorded the worst decline — 36 percentage points in a single year

3.8 Regional Analysis

Five-region analysis reveals a clear development hierarchy:

Coastal Karnataka (3 districts):

Average DDI: 63.24 | Average Income: ₹5.07L

Highest performing region across all indicators

Malnad Karnataka (2 districts):

Average DDI: 51.80 | Average Income: ₹4.42L

Second highest — driven by tourism and forest economy

South Karnataka (11 districts):

Average DDI: 47.59 | Average Income: ₹3.35L

Moderate development — includes Bengaluru satellites and Mysuru

Central Karnataka (2 districts):

Average DDI: 32.41 | Average Income: ₹2.35L

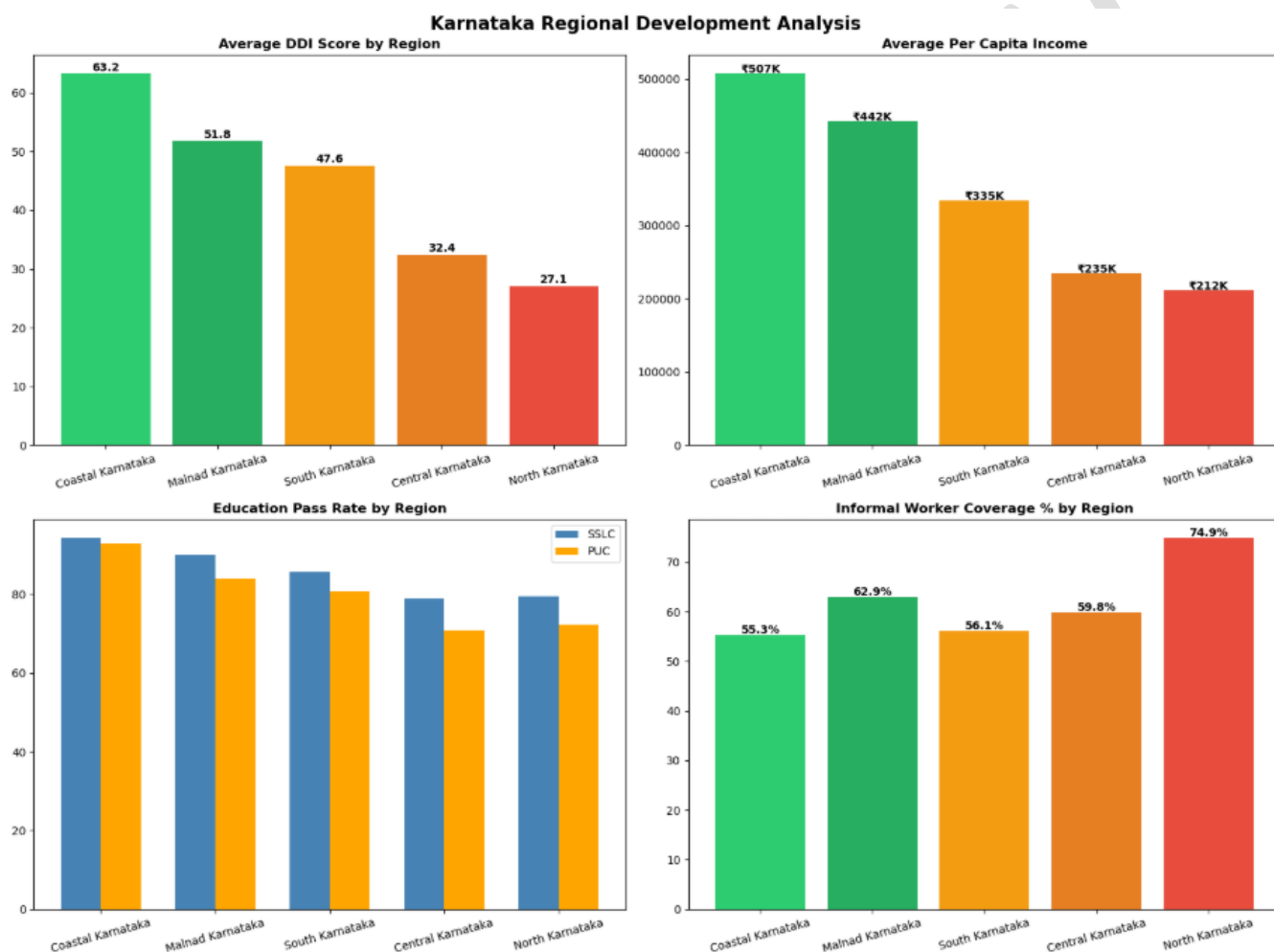
Below average — Davanagere and Chitradurga

North Karnataka (13 districts):

Average DDI: 27.05 | Average Income: ₹2.12L

Lowest performing region — 13 of Karnataka's most underdeveloped districts

Coastal Karnataka's average DDI of 63.24 is more than double North Karnataka's 27.05, confirming a systematic regional development gap that transcends individual district performance.



Key Fact: Coastal Karnataka average DDI (63.24) is more than double North Karnataka average DDI (27.05)

3.9 Gini Coefficient — Inequality Analysis

Gini coefficients reveal unequal distribution of development indicators:

DDI Score = 0.2151 (Moderate)

Per Capita Income = 0.2193 (Moderate)

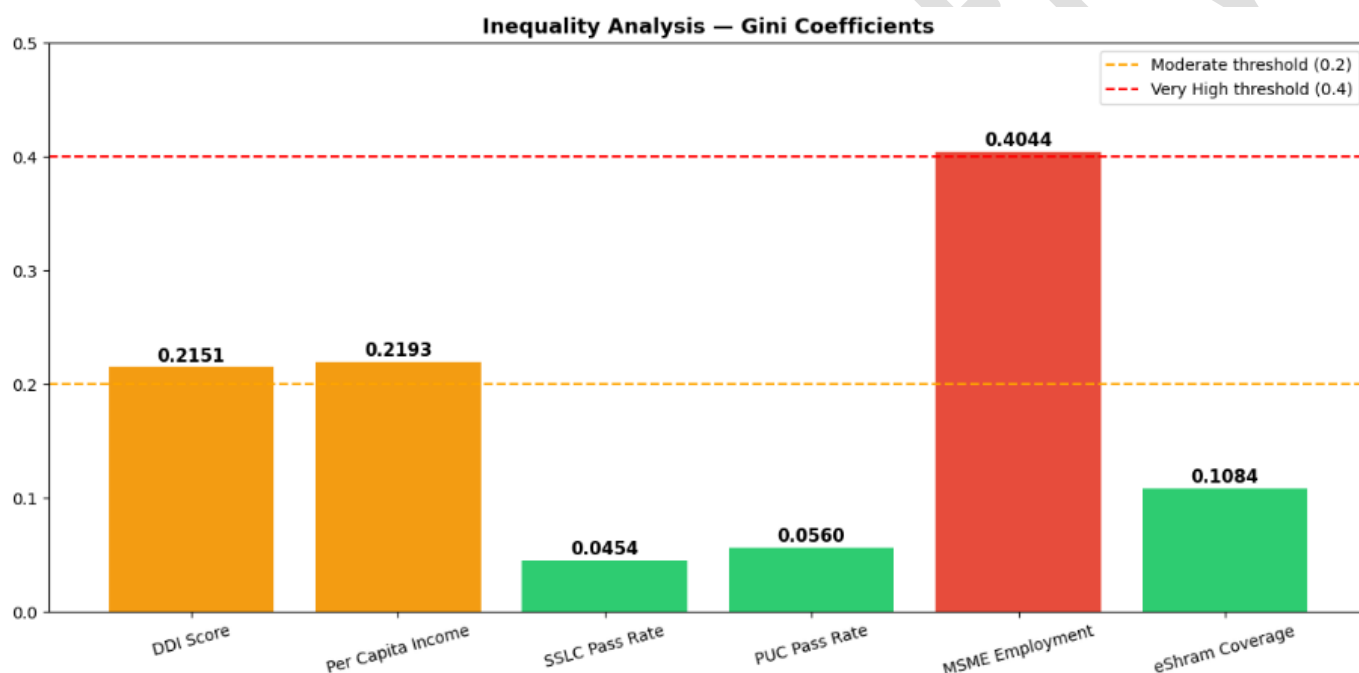
SSLC Pass Rate = 0.0454 (Low)

PUC Pass Rate = 0.0560 (Low)

MSME Employment = 0.4044 (Very High)

eShram Coverage = 0.1084 (Low)

Key insight: Educational access shows low inequality (Gini 0.04-0.05) across Karnataka districts. However employment through MSMEs shows very high inequality (Gini 0.40), indicating that job creation is heavily concentrated in urban centers while rural districts have comparable education but far fewer formal employment opportunities.



Key Fact: SSLC education Gini = 0.04 (Low) but MSME Employment Gini = 0.40 (Very High) — education reaches everywhere but jobs do not.

3.10 Sensitivity Analysis

DDI rankings remain consistent across three weight scenarios:

Original Weights:

Top 3: Bengaluru Urban (79.79), Dakshina Kannada (75.23), Udupi (69.80)

Bottom 3: Raichur, Yadagiri, Kalaburagi

Education Heavy Weights:

Top 3: Bengaluru Urban, Dakshina Kannada, Udupi

Bottom 3: Raichur, Yadagiri, Kalaburagi

Economy Heavy Weights:

Top 3: Bengaluru Urban, Dakshina Kannada, Udupi

Bottom 3: Raichur, Yadagiri, Kalaburagi

The top 3 and bottom 3 districts remain

identical across all weight scenarios confirming that the District Development Index is robust and not sensitive to weight assumptions.

ANALYSIS 4 – SENSITIVITY ANALYSIS

District	Original Weights	Education Heavy	Economy Heavy
Bengaluru Urban	79.787038	78.069428	87.810958
Dakshina Kannada	75.233050	80.647508	70.861560
Udupi	69.800670	75.658660	64.770871
Kodagu	54.768722	61.569403	41.354603
Chikkamagaluru	54.400061	56.527251	49.468358
Bengaluru Rural	51.213347	55.221379	43.096527
Hassan	49.749909	55.530838	41.113700
Shivamogga	49.206512	55.364930	42.836811
Mysuru	45.415444	50.354805	35.150075
Uttara Kannada	44.694276	53.509384	33.706434
Tumakuru	43.202244	43.210752	36.051625
Mandya	42.735230	45.083739	35.313921
Kolar	41.399518	46.451228	29.601448
Ramanagara	39.785457	42.992495	32.473620
Chamarajanagar	39.064368	42.153343	29.573287
Bagalkote	37.878304	43.442400	31.487315
Chikkaballapur	36.404142	39.130043	26.144206
Dharwad	35.838827	37.946299	29.516392
Davanagere	35.362057	38.404059	26.848328
Ballari	34.046353	33.286871	30.994252
Gadag	32.281360	34.587934	22.339452
Vijayanagar	31.186132	37.395084	19.603161
Chitradurga	29.456212	27.682925	22.451147
Vijayapura	28.394134	34.853030	19.736223
Haveri	28.025930	31.719684	19.201686
Koppal	27.876180	30.491126	17.144980
Bidar	25.915342	28.162246	19.252951
Belagavi	25.828793	32.278912	21.576983
Raichur	19.029400	20.818080	12.804579
Yadagiri	15.545380	11.383762	7.899502
Kalaburagi	9.824761	8.839761	6.497268

Key Fact: Top 3 and bottom 3 districts remain identical across all weight scenarios confirming DDI robustness

3.11 Future Prediction Scenarios

Predictive modeling for bottom 5 districts reveals the relative impact of different policy interventions:

Kalaburagi (Current DDI: 9.82):

10% Income Increase = DDI: 14.88 (+5.06)
 10% Education Improvement = DDI: 20.4 (+10.58)
 Combined Improvement = DDI: 20.85 (+10.03)

Yadagiri (Current DDI: 15.55):

10% Income Increase = DDI: 20.75 (+5.20)
 10% Education Improvement = DDI: 21.58 (+6.03)
 Combined Improvement = DDI: 24.6 (+9.05)

Raichur (Current DDI: 19.03):

10% Income Increase = DDI: 24.06 (+5.03)
 10% Education Improvement = DDI: 24.32 (+5.29)
 Combined Improvement = DDI: 27.74 (+8.71)

Key finding: Education improvement consistently yields approximately twice the development impact compared to equivalent income growth for Karnataka's most underdeveloped districts.

ANALYSIS 5 – FUTURE PREDICTION SCENARIOS

Kalaburagi (Current DDI: 9.82)		
10% Income Increase	→	14.88 (+5.06)
10% Education Improvement	→	20.4 (+10.58)
Combined Improvement	→	20.85 (+11.03)
Yadagiri (Current DDI: 15.55)		
10% Income Increase	→	20.75 (+5.20)
10% Education Improvement	→	21.58 (+6.03)
Combined Improvement	→	24.6 (+9.05)
Raichur (Current DDI: 19.03)		
10% Income Increase	→	24.06 (+5.03)
10% Education Improvement	→	24.32 (+5.29)
Combined Improvement	→	27.74 (+8.71)
Belagavi (Current DDI: 25.83)		
10% Income Increase	→	25.64 (+/-0.19)
10% Education Improvement	→	33.51 (+7.68)
Combined Improvement	→	34.06 (+8.23)
Bidar (Current DDI: 25.92)		
10% Income Increase	→	27.34 (+1.42)
10% Education Improvement	→	31.57 (+5.65)
Combined Improvement	→	32.33 (+6.41)

Key Fact: 10% education improvement gives 2x more DDI impact than 10% income increase

4. Key Findings

This study identifies eight key findings from the analysis of 31 Karnataka districts:

Finding 1 — Extreme Development Inequality:

Bengaluru Urban scores 79.79 on the DDI while Kalaburagi scores only 9.82 — a gap of nearly 70 points within the same state. This represents one of the most extreme intra-state development divides in India.

Finding 2 — 65% of Karnataka is Underdeveloped:

K-Means clustering reveals that 20 of 31 districts fall in the Underdeveloped tier. Only 1 district achieves Highly Developed status. Karnataka's development story is essentially the story of one city — Bengaluru Urban.

Finding 3 — Coastal Karnataka Consistently Leads:

Udupi and Dakshina Kannada rank 2nd and 3rd on the DDI despite having smaller economies than Bengaluru. Their consistent performance across income, education, and employment indicators suggests a well-rounded development model driven by trade, tourism, and education.

Finding 4 — North Karnataka is Systematically Left Behind:

13 of Karnataka's 31 districts are in North Karnataka. This region has the lowest average DDI (27.05), lowest average income (₹2.12L), highest informal worker coverage (74.89%), and highest youth unemployment (6,212 per district). This is not coincidence, it is a systemic regional development failure.

Finding 5 — Income Drives 60.6% of Development:

Random Forest feature importance analysis reveals that Per Capita Income alone explains 60.6% of development variation across districts. Education explains another 31.9%. Together income and education explain 92.5% of district development in Karnataka.

Finding 6 — Education is Equal but Jobs are Not:

Gini coefficient analysis reveals that educational access is relatively equal across Karnataka (SSLC Gini 0.04, PUC Gini 0.06). However MSME employment is highly concentrated (Gini 0.40). Karnataka has successfully spread education but failed to spread formal employment opportunities beyond urban centers.

Finding 7 — Universal Education Decline in 2025:

Time series analysis reveals that 28 of 31 districts declined in SSLC pass rates between 2024 and 2025. All 31 districts declined in PUC pass rates. Vijayapura recorded the most alarming decline PUC pass rate fell by 36 percentage points in a single year.

Finding 8 — Education Investment Outperforms Income Investment:

Predictive modeling reveals that a 10% improvement in education indicators yields approximately twice the development impact compared to equivalent income growth for Karnataka's most underdeveloped

districts. This has profound implications for government resource allocation decisions.

5. Policy Recommendations

Based on the findings of this study, six policy recommendations are presented:

Recommendation 1 — Prioritize North Karnataka:

The Karnataka government should designate North Karnataka as a Special Development Zone and redirect a minimum of 40% of state development funds to the 13 North Karnataka districts. Current funding patterns appear to reinforce existing inequalities rather than address them.

Recommendation 2 — Emergency Education Intervention:

The universal decline in education pass rates between 2024 and 2025 demands immediate investigation and intervention. Special remedial education programs should be launched in the bottom 10 districts particularly Kalaburagi, Yadagiri, Vijayapura, and Raichur where both SSLC and PUC declines are most severe.

Recommendation 3 — Formalize the Informal Economy:

North Karnataka has an average informal worker coverage of 74.89% the highest in the state. Skill development programs, MSME registration drives, and formal employment creation schemes should be urgently implemented in this region to convert informal workers into formal economy participants.

Recommendation 4 — MSME Decentralization:

MSME employment shows very high inequality (Gini 0.40) heavily concentrated in Bengaluru Urban. Government industrial policies should incentivize MSME establishment in underdeveloped districts through tax benefits, infrastructure investment, and simplified registration processes.

Recommendation 5 — Adopt DDI as Annual Monitoring Tool:

The District Development Index developed in this study should be adopted by the Karnataka government as an annual monitoring framework. Tracking DDI scores year-on-year will enable evidence-based assessment of whether government interventions are actually reducing regional inequality.

Recommendation 6 — Replicate the Coastal Karnataka Model:

Coastal Karnataka achieves the highest average DDI (63.24) with a balanced combination of trade, education, tourism, and healthcare. A detailed study of the Coastal Karnataka development model should be conducted and its successful elements replicated in underdeveloped regions.

6. Conclusion

This study presents a comprehensive data-driven analysis of development across Karnataka's 31 districts using seven government datasets, four machine learning models, and five advanced analytical techniques.

The District Development Index constructed in this study provides a robust, multidimensional measure of district-level development that goes beyond single-indicator approaches. Sensitivity analysis confirms that the index rankings are stable across different weight assumptions validating its reliability as a policy tool.

The findings paint a stark picture of Karnataka's development landscape. While Bengaluru Urban and the coastal districts of Dakshina Kannada and Udupi have achieved impressive development outcomes, 65% of Karnataka's districts remain underdeveloped. North Karnataka home to 13 of the state's 31 districts consistently underperforms across every indicator analyzed in this study.

Perhaps the most important finding of this study is the education-employment paradox. Karnataka has made significant progress in spreading educational access SSLC and PUC pass rates show relatively equal distribution across districts (Gini 0.04-0.06). However formal employment opportunities remain heavily concentrated in urban centers (MSME Employment Gini 0.40). Young people in North Karnataka are receiving education but not finding formal employment driving the high youth unemployment and informal worker registration rates observed in the region.

Machine learning models confirm that income and education together explain 92.5% of district development variation. Predictive scenario analysis demonstrates that education investment yields twice the development impact of equivalent income transfers providing a clear evidence base for government resource allocation decisions.

The universal decline in education performance between 2024 and 2025 is a finding that demands immediate government attention. A statewide decline of this magnitude affecting 28 of 31 districts in SSLC and all 31 districts in PUC represents a serious threat to Karnataka's human capital development.

In conclusion, Karnataka cannot achieve its full economic potential while 65% of its districts remain underdeveloped. Targeted, evidence-based policy interventions focusing on education quality, formal employment creation, and MSME decentralization in North Karnataka are urgently needed. The District Development Index developed in this study provides the analytical foundation for such interventions.

7. Limitations

This study acknowledges the following limitations:

Limitation 1 — Small Sample Size: With only 31 districts the dataset is relatively small for machine learning applications. Results should be interpreted with caution and validated with larger datasets when available.

Limitation 2 — Data Imputation: Ramanagara district was missing from the eShram dataset and was imputed using column mean values. This imputation may not accurately reflect the district's actual informal worker coverage.

Limitation 3 — Education Data Verification: The observed universal decline in education pass rates between 2024 and 2025 requires verification with official KSEAB data before drawing definitive policy conclusions. Changes in examination patterns or evaluation criteria may partially explain this trend.

Limitation 4 — Weight Subjectivity: Although sensitivity analysis confirms ranking stability, the weights assigned to each indicator in the DDI involve a degree of subjectivity. Different stakeholders may prioritize different indicators.

Limitation 5 — Temporal Limitation: This study uses data from 2024-26 only. A longitudinal analysis across multiple years would provide a more accurate picture of development trends.

Limitation 6 — Missing Indicators: Important development indicators such as healthcare infrastructure, crime rates, agricultural productivity, and road connectivity were not included due to data unavailability at the time of study.

8. References

1. Government of Karnataka (2026). Economic Survey of Karnataka 2025-26. Department of Planning, Programme Monitoring and Statistics, Bengaluru.
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4. Government of Karnataka (2025). District-wise Number Registered under eShram Portal 2025.

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5. Government of Karnataka (2025). District-wise Number Passed SSLC 2024 and 2025.

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6. Government of Karnataka (2025). District-wise PUC Pass Percentage 2024 and 2025.

Source: data.opencity.in

Appendix A — Technical Details

Programming Language = Python 3.12

Development Environment = Jupyter Notebook

Key Libraries Used = Pandas, NumPy, Scikit-learn, Statsmodels, Matplotlib, Seaborn, SQLite3

Appendix B — Dataset Summary

Total datasets used = 7

Total districts = 31

Total indicators = 22

Missing values = 0

Time period = 2024-2026

Appendix C — Model Summary

K-Means Clustering = K=3, Elbow method

Random Forest Reg = 100 estimators, $R^2=0.98$

Random Forest Clf = 100 estimators, Accuracy=100%

OLS Regression = All p-values < 0.001

Appendix D — Files

Karnataka_DDI_Final.ipynb = Python notebook

master_dataset_final.csv = Final dataset

karnataka_powerbi.xlsx = Power BI data

karnataka_development.db = SQL database

15 visualization PNG files